[GATE-EC-2008]

(22) The two numbers represented in signed 2's complement form are

P = 11101101 and Q = 11100110. If Q is subtracted from P, the value obtained in signed 2's complement form is

(A) 100000111

(B) 00000111

(C) 11111001

(D) 111111001

[GATE-EC-2001]

(23) The 2's complements representation of -17 is

(A) 101110 (B) 101111

(C) 111110 (D) 110001

[GATE – IN – 2018]

(24) The representation of the decimal number $(27.625)_{10}$ in base-2 number system is

(A) 11011.110 (B) 11101.101

(C) 11011.101 (D) 10111.110

The Boolean expression

$$F(X,Y,Z)=\bar{X}Y\bar{Z}+X\bar{Y}\bar{Z}+XY\bar{Z}+XYZ$$

Converted into canonical product of sum (POS) form is :

(A) $(X+Y+Z)(X+Y+\bar{Z})$

$(X+\bar{Y}+\bar{Z})(\bar{X}+Y+\bar{Z})$

(B) $(X+\bar{Y}+Z)(\bar{X}+Y+\bar{Z})$

$(\bar{X}+\bar{Y}+Z)(\bar{X}+\bar{Y}+\bar{Z})$

(C) $(X+Y+Z)(\bar{X}+Y+\bar{Z})$

$(X+\bar{Y}+Z)(\bar{X}+\bar{Y}+\bar{Z})$

(D) $(X+\bar{Y}+\bar{Z})(\bar{X}+Y+Z)$

$(\bar{X}+\bar{Y}+Z)(X+Y+Z)$

[GATE-EC-2015]

A function of Boolean variables, X, Y and Z is expressed in terms of the main-terms as $F(X, Y, Z) = \Sigma\ (1, 2, 5, 6, 7)$

Which one of the product of sums given below is equal to the function $F(X, Y, Z)$?

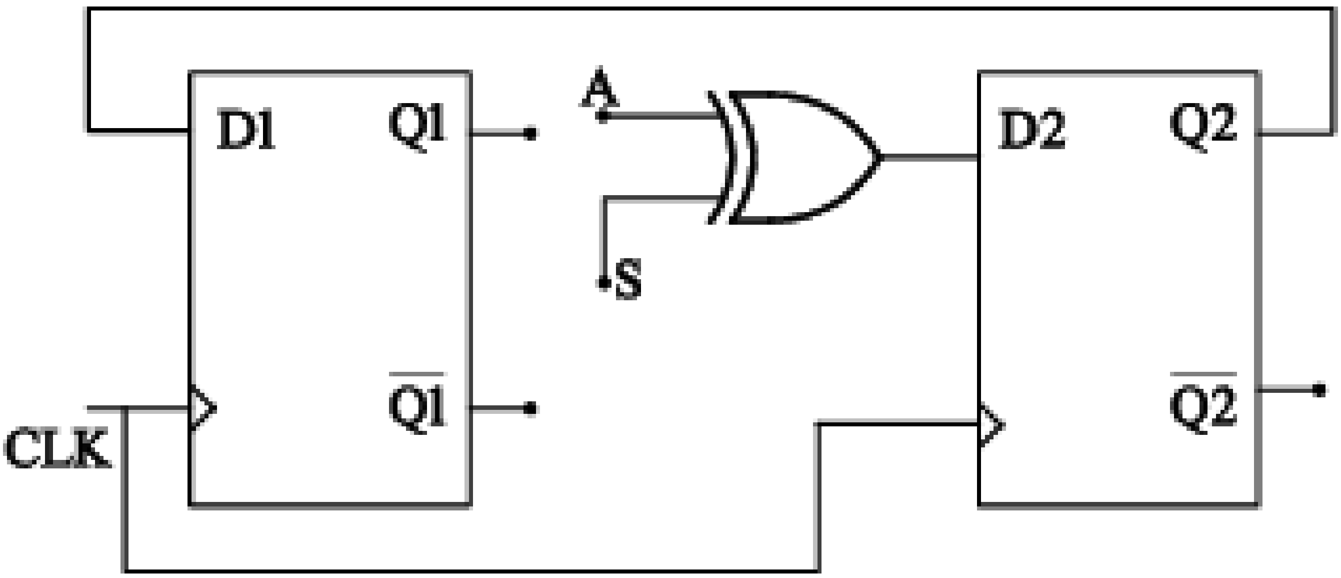
(A) $(\bar{X}+\bar{Y}+\bar{Z})\cdot(\bar{X}+Y+Z)\cdot$

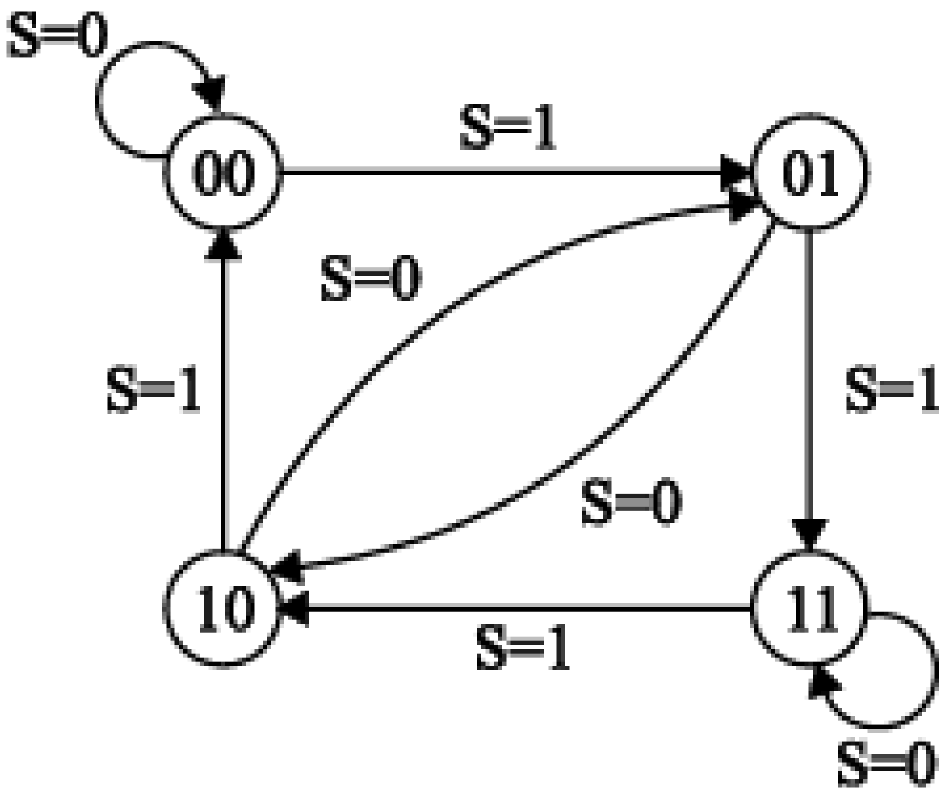
$(X+\bar{Y}+\bar{Z})$

(B) $(X+Y+Z)\cdot(X+\bar{Y}+\bar{Z})\cdot$

$(\bar{X}+Y+Z)$

(60) The digital logic shown in the figure satisfies the given state diagram when Q1 is connected to input A of the XOR gate. Suppose the XOR gate is replaced an by an XNOR gate. Which one of the following options preserves the state diagram?





If the functions W, X, Y and Z are as follows

$$W = R + \bar{P}Q + \bar{R}S$$

$$X = PQ\bar{R}\bar{S} + \bar{P}\bar{Q}\bar{R}\bar{S} + P\bar{Q}\bar{R}\bar{S}$$

$$Y = RS + \overline{PR + P\bar{Q} + \bar{P}\bar{Q}}$$

$$Z = R + S + \overline{PQ + \bar{P}\bar{Q}\bar{R} + P\bar{Q}\bar{S}}$$

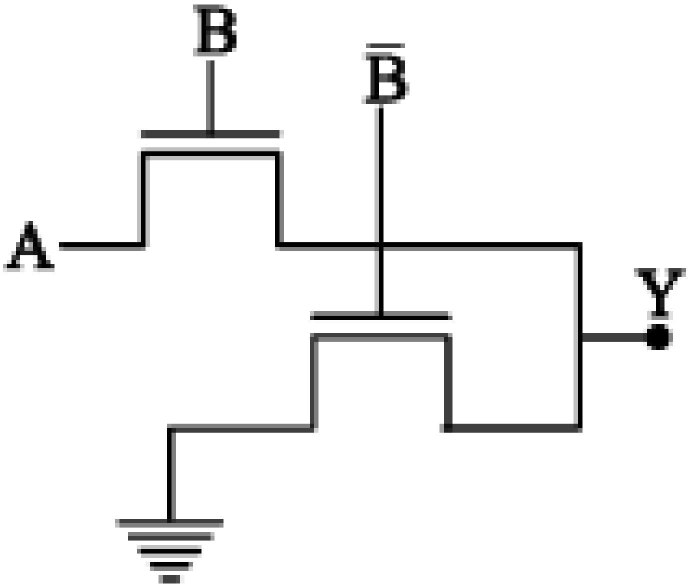
(A) $W = Z, X = \bar{Z}$

(B) $W = Z, X = Y$

(C) $W = Y$

(D) $W = Y = \bar{Z}$

(116) The logic functionality realized by the circuit shown below is :

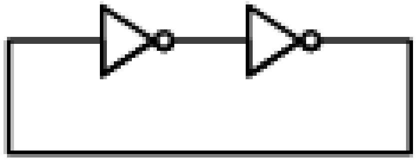


(A) OR **(B)** XOR

(C) NAND **(D)** AND

[GATE–S4–EC–2016]

(11) The digital circuit using two inverters shown in Fig. will act as



(A) A bistable multi-vibrator

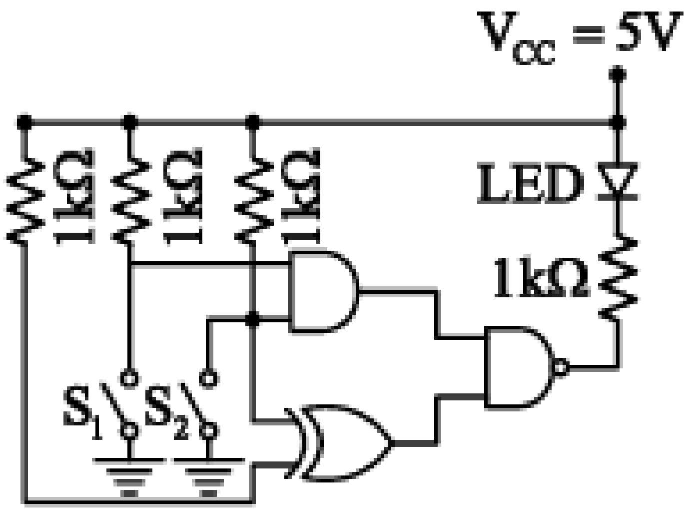
(B) An astable multi-vibrator

(C) A Monostable multi-vibrator

(D) An oscillator

[GATE-EC-2001]

(12) In the figure, the LED



(A) emits light when both S_1 and S_2 are closed.

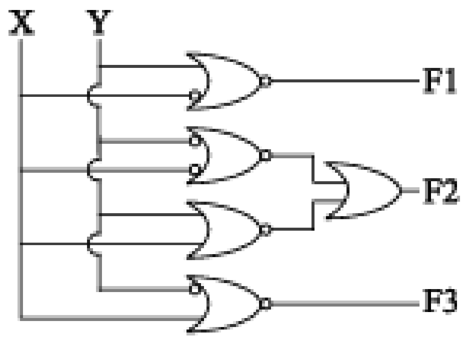
(B) emits light when both S_1 and S_2 are open.

(C) emits light when only of S_1 and S_2 is closed.

(D) does not emit light, irrespective of the switch positions.

[GATE-EC-2015]

The circuit shown in the given figure is :



(A) An adder circuit

(B) A subtractor circuit

(C) A comparator circuit

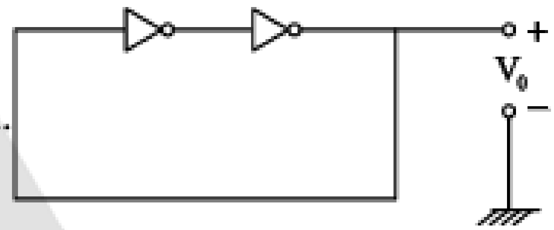
(D) A parity generator circuit

[IES – EC – 2002]

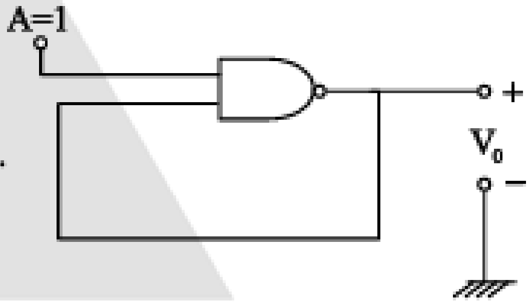
Consider the following circuits (Assume all gates to have a finite propagation delay)

Which of these circuits generate a periodic square wave output?

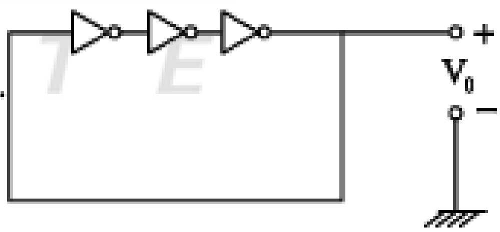
1.



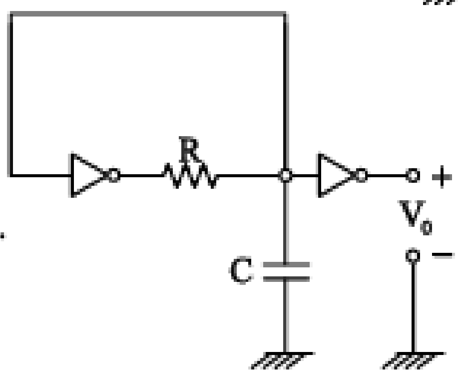
2.



3.



4.



(A) 1 and 2

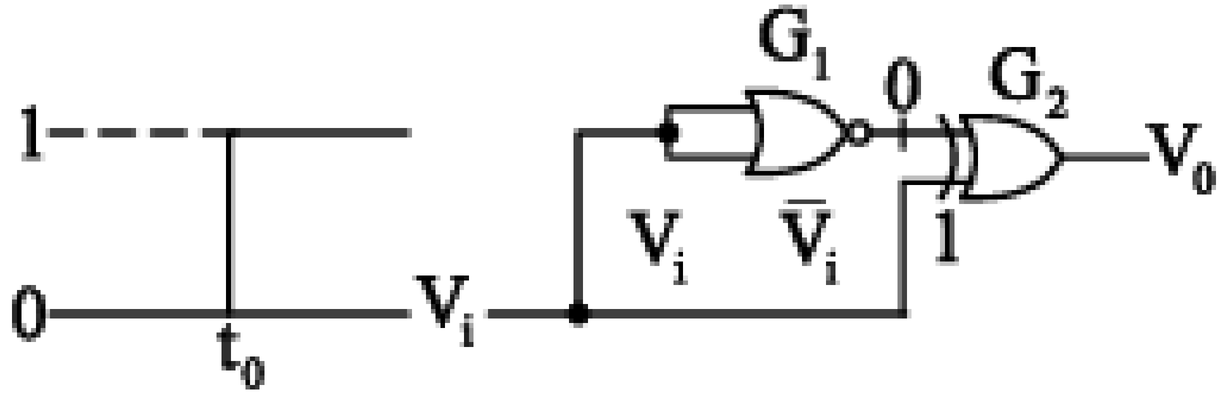
(B) 3 and 4

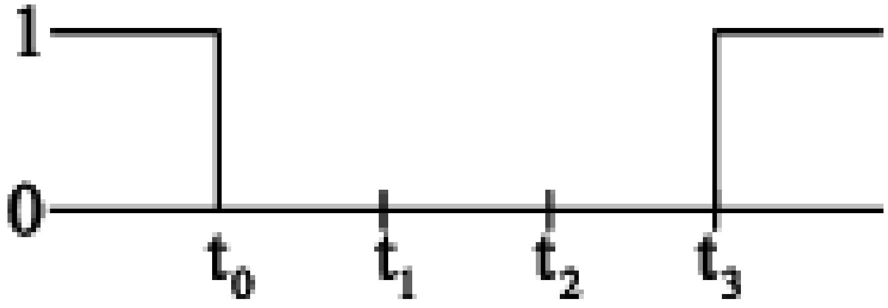
(C) 2, 3 and 4

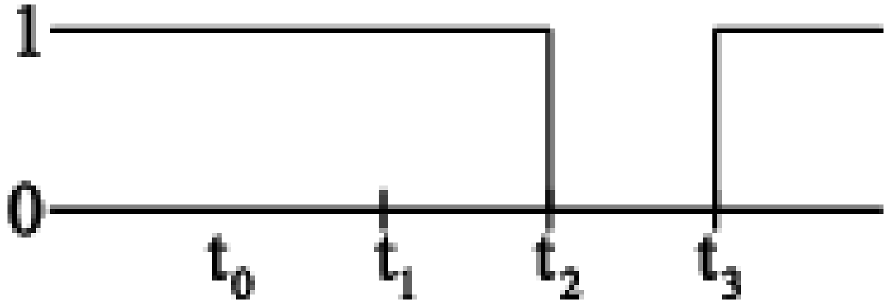
(D) 1, 2, 3 and 4

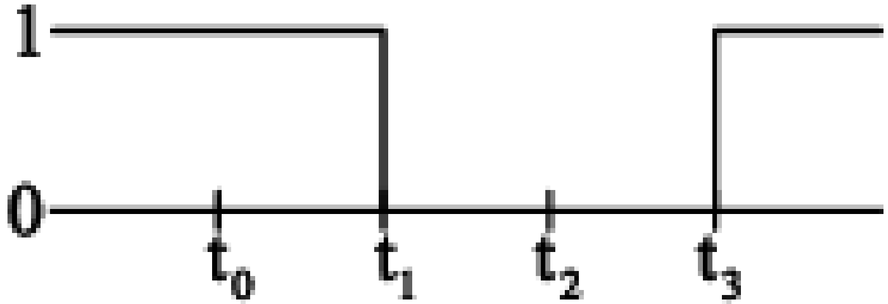
[GATE EC 2002]

The gates G_1 and G_2 in the figure have propagation delays of 10n sec and 20n sec respectively. If the input V_i makes an abrupt change from logic 0 to 1 at time t = t_0, then the output waveform V_0 is :



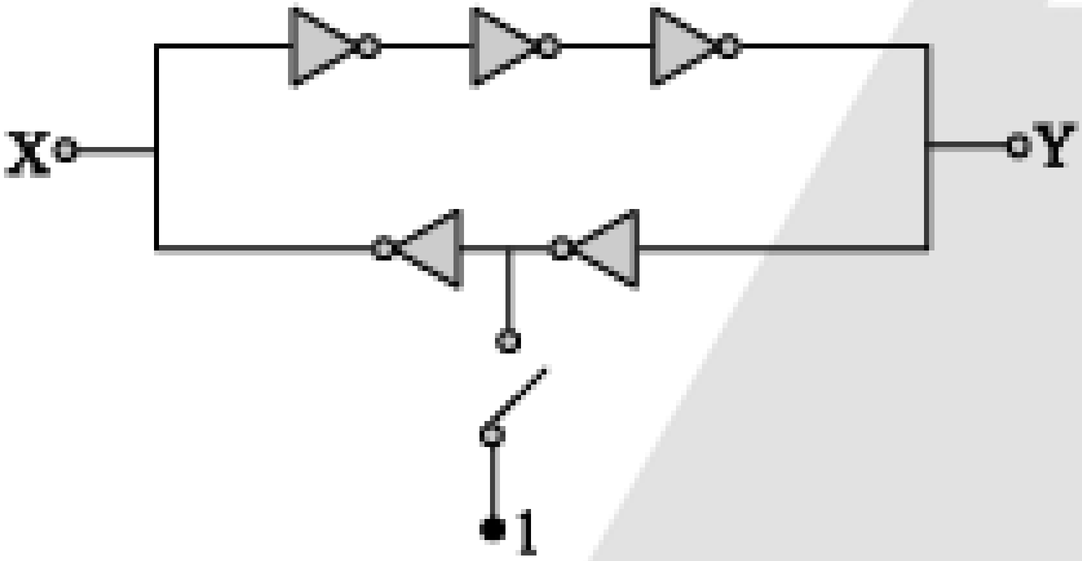
(A)


(B)


(C)


[GATE - IN - 2015]

In the circuit shown, the switch is momentarily closed and then opened. Assuming the logic gates to have equal non-zero delay, at steady state, the logic states of X and Y are



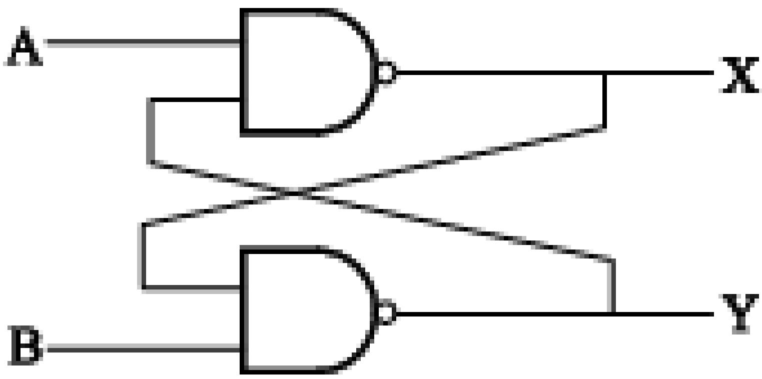
(A) X is latched, Y toggles continuously

(B) X and Y are both latched

(C) Y is latched, X toggles continuously

(D) X and Y both toggle continuously

(103) In below figure, A = 1 and B = 1. The input B is now replaced by a sequence 101010 .... the outputs x and y will be



(A) fixed at 0 and 1, respectively

(B) x = 1010 ------- while y = 0101 ----

(C) x = 1010 ------- and y = 1010 -------

(D) fixed at 1 and 0, respectively

RS \ PQ	00	01	11	10
00	0	0	0	0
01	1	0	0	1
11	1	0	0	1
10	0	0	0	0

X=0

RS \ PQ	00	01	11	10
00	0	1	1	0
01	0	0	0	0
11	0	0	0	0
10	0	1	1	0

X=1

(A) $\overline{P}\,\overline{Q}S\overline{X} + P\overline{Q}S\overline{X} + Q\overline{R}\,\overline{S}X + QR\overline{S}X$

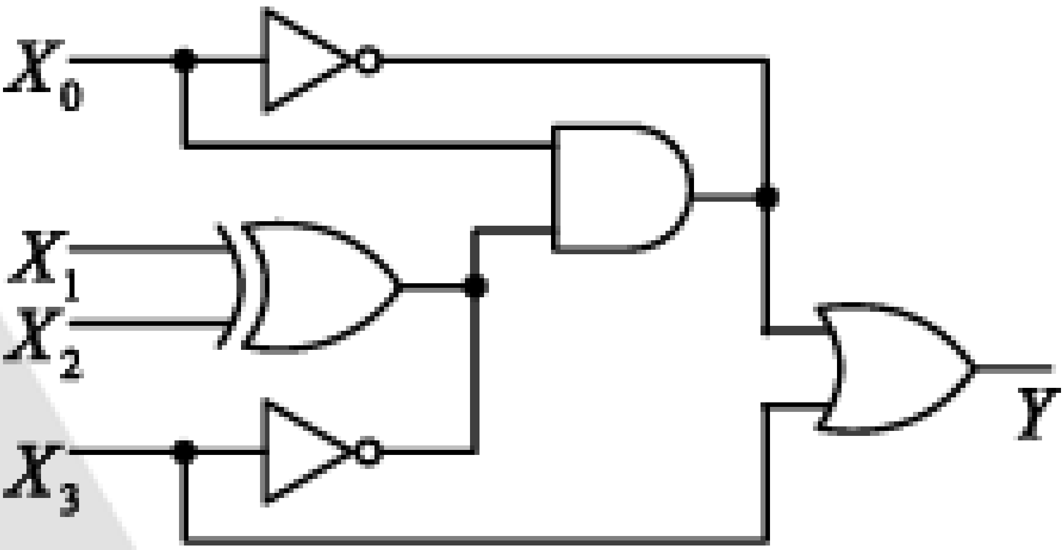
(B) $\overline{Q}S\overline{X} + Q\overline{S}X$

(C) $\overline{Q}SX + Q\overline{S}X$

(D) $\overline{Q}S + Q\overline{S}$

[GATE–S1–EC–2016]

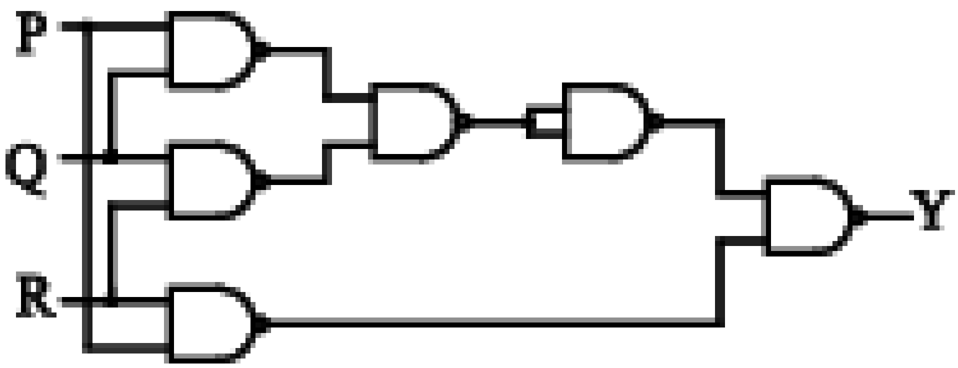
) The logic gates shown in the digital circuit below use strong pull-down nMOS transistors for LOW logic level at the outputs. When the pull-downs are off, high-value resistors set the output logic levels to HIGH (i.e. the pull-ups are weak). Note that some nodes are intentionally shorted to implement "wired logic". Such shorted nodes will be HIGH only if the outputs of all the gates whose outputs are shorted are HIGH.



The number of distinct values of $X_3X_2X_1X_0$ (out of the 16 possible values) that give $Y = 1$ is ________.

[GATE – EE – 2018]

The output Y in the circuit below is always "1" when



(A) two or more of the inputs P, Q, R are "0"

(B) two or more of the inputs P, Q, R are "1"

(C) any odd number of the inputs P, Q, R is "0"

(D) any odd number of the inputs P, Q, R is "1"

A digital multiplexer can be used for which of the following?

1. Parallel to serial conversion
2. Many-to-one switch
3. To generate memory chip select
4. For code conversion

Select the correct answer using the code given below:

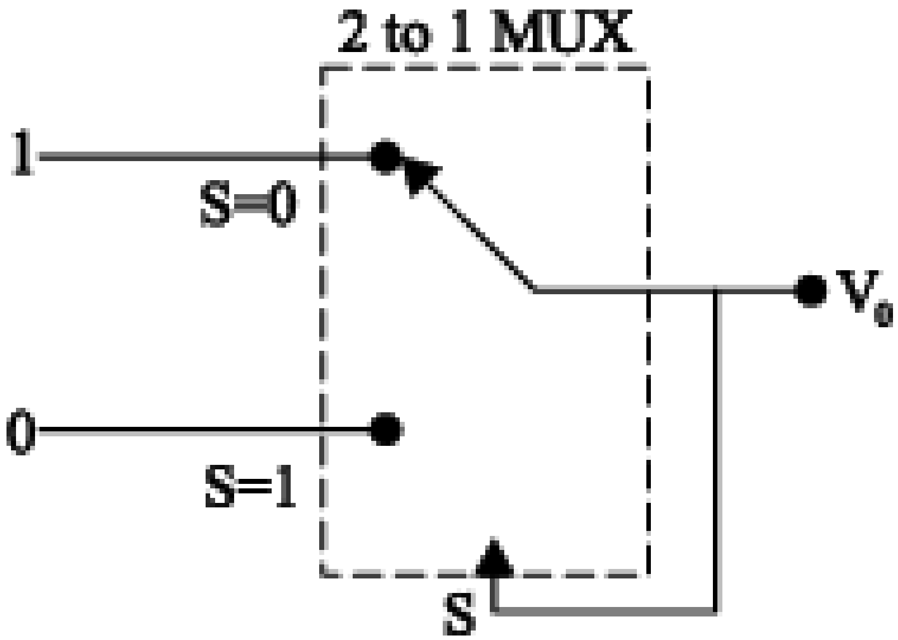
(A) 1, 3 and 4

(B) 2, 3 and 4

(C) 1 and 2 only

(D) 2 and 3 only

A 2-to 1 digital multiplexer having a switching delay of 1 μs is connected as shown in Fig. The output of the multiplexer is tied to its own select input S. The inputs which gets selected when S = 0 is tied to 1 and the input that that go is selected when S = 1 is tied to 0. The output V_0 will be



(A) 0

(B) 1

(C) pulse train of frequency 0.5 MHz

(D) pulse train of frequency 1.0 MHz

[GATE - IN - 2005]

Which one of the following statements is not correct?

(A) An 8 input MUX can be used to implement any 4 variable function

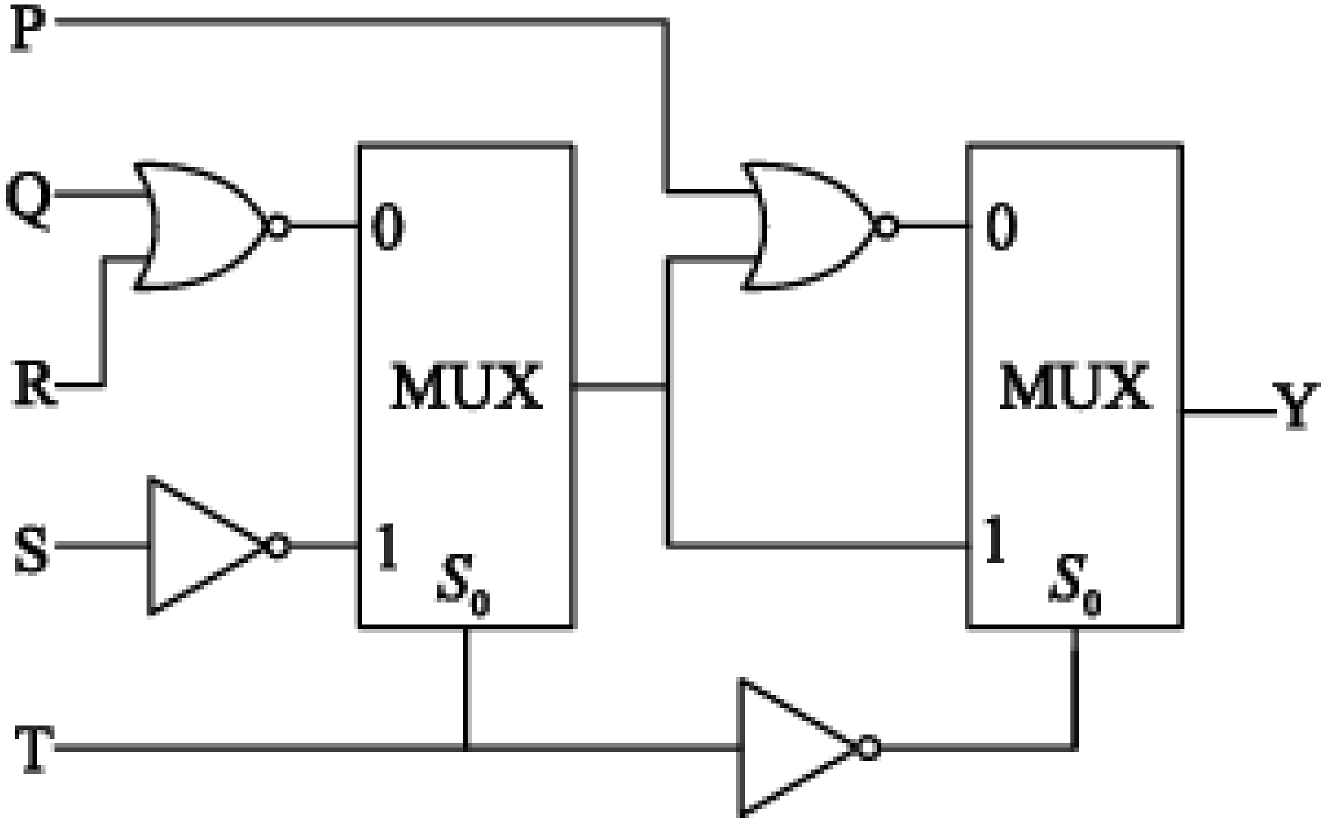
(B) A 3 line to 8 line DEMUX can be used to implement any 4 variable function

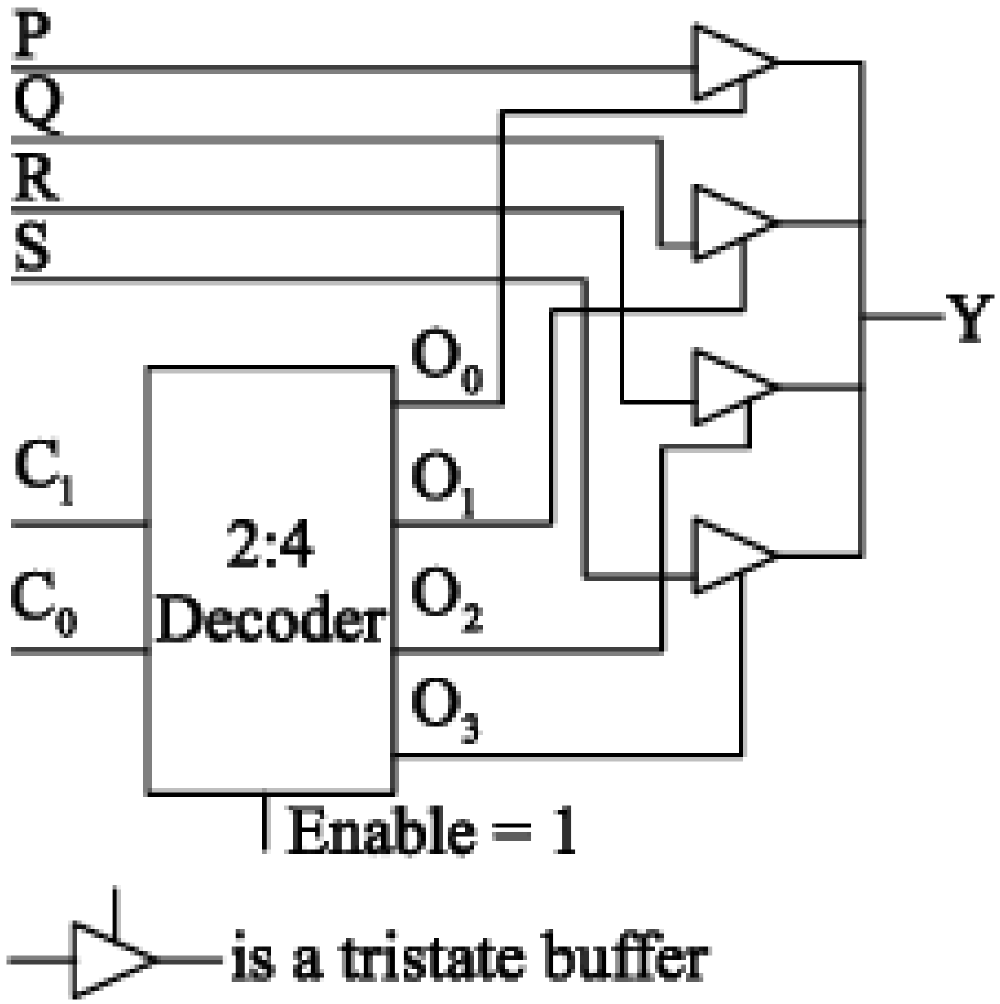
(C) A 64 input MUX can be built nine 8 input MUXs

(D) A 6 line to 64 line DEMUX can be built using nine 3 line to 8 line DEMUXs

[IES – EC – 1997]

(46) For the circuit shown in the figure, the delays of NOR gates, multiplexers and inverters are 2 ns, 1.5 ns and 1 ns, respectively. If all the inputs P, Q, R, S and T are applied at the same time instant, the maximum propagation delay (in ns) of the circuit is ________





(A) 2-to-1 multiplexer

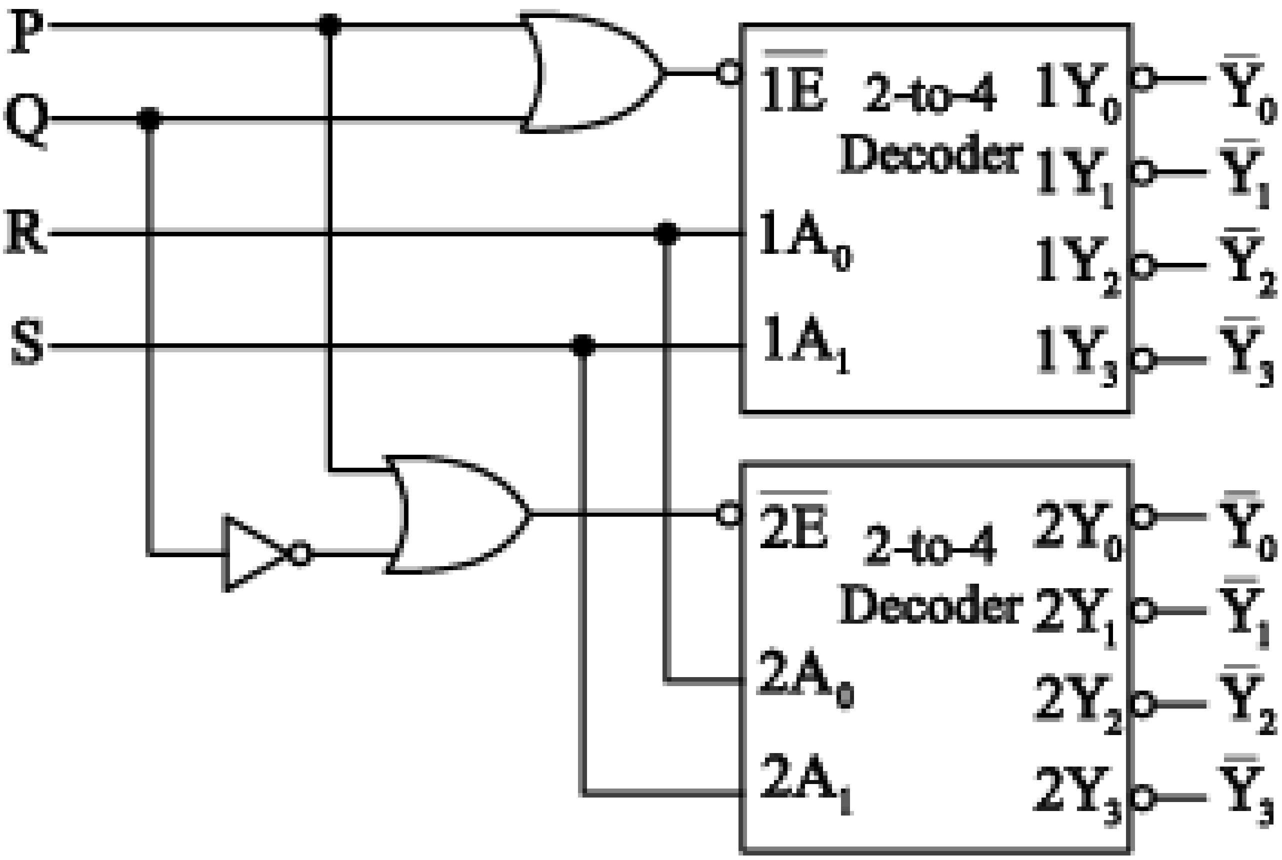
(B) 4-to-1 multiplexer

(C) 7-to-1 multiplexer

(D) 6-to-1 multiplexer

[GATE–S2–EC–2017]

(1) A 1-to-8 demultiplexer with data input D_{in}, address inputs S_0, S_1, S_2 (with S_0 as the LSB) and $\overline{Y}_0$ to $\overline{Y}_7$ as the eight demultiplexed outputs, is to be designed using two 2-to-4 decoders (with enable input $\overline{E}$ and address inputs A_0 and A_1) as shown in the figure. D_{in}, S_0, S_1 and S_2 are to be connected to P,Q,R and S , but not necessarily in this order. The respective input connections to P,Q,R and S terminals should be :

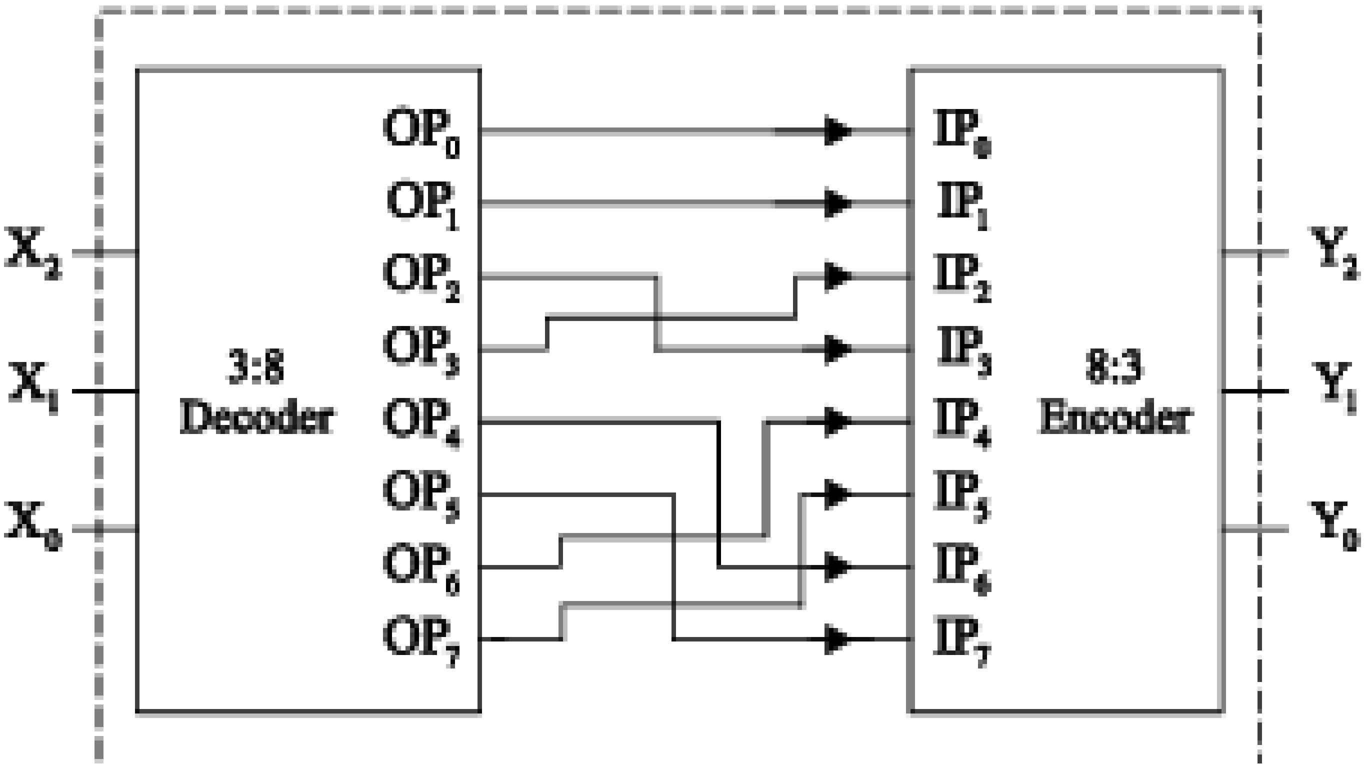


(A) S_2, D_{in}, S_0, S_1 **(B)** S_1, D_{in}, S_0, S_2

(C) D_{in}, S_0, S_1, S_2 **(D)** D_{in}, S_2, S_0, S_1

[GATE–S1–EC–2016]

(8) Identify the circuit below.



(A) Binary to Gray code converter

(B) Binary to XS3 converter

(C) Gray to Binary converter

(D) XS3 to Binary converter

Consider the following statements:

A 4:16 decoder can be constructed (with enable input) by

1. Using four 2:4 decoders (each with an enable input) only.
2. Using five 2:4 decoders (each with an enable input) only.
3. Using two 3:8 decoders (each with an enable input) only.
4. Using two 3:8 decoders (each with an enable input) and an inverter.

Which of the statements given above is/are correct ?

(A) 1 only **(B)** 1 only

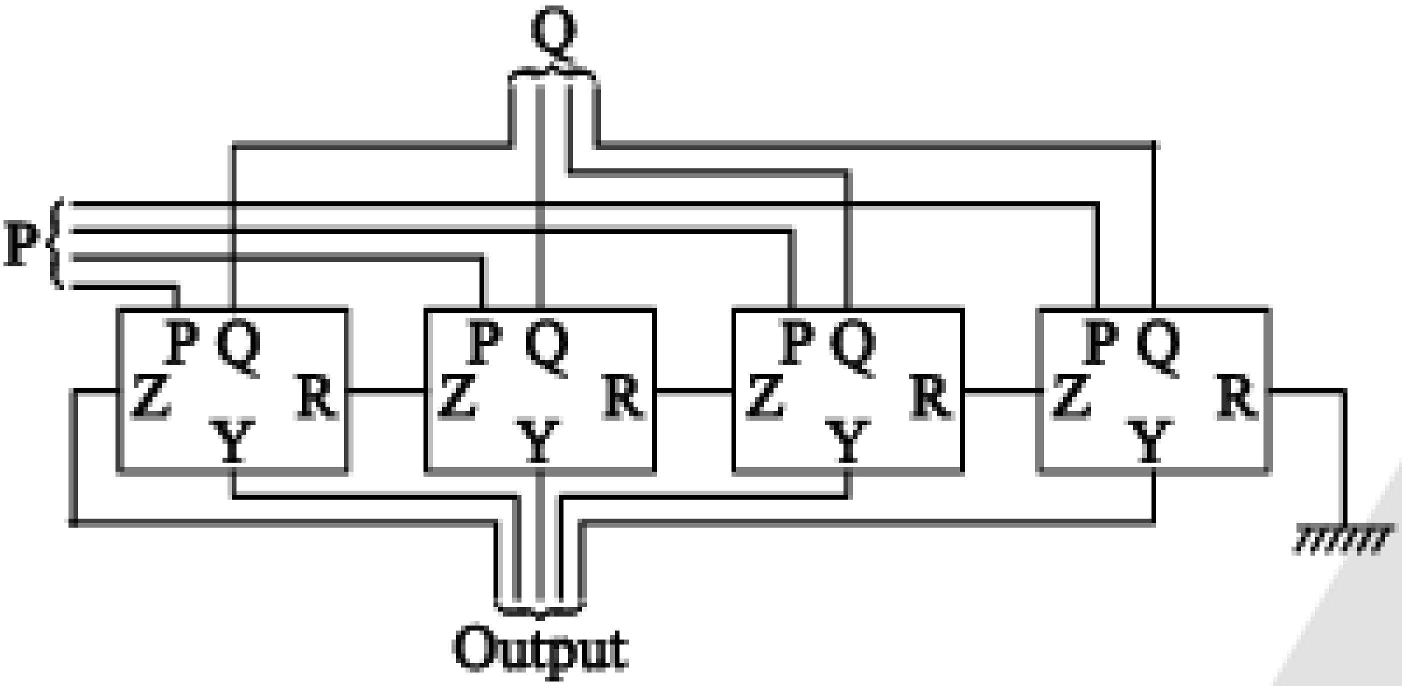
(C) 2 and 4 **(D)** None

[GATE - EE - 2008]

(16) The circuit shown in the figure has 4 boxes each described by inputs P, Q, R and outputs Y, Z with

$$Y = P \oplus Q \oplus R$$

$$Z = RQ + \overline{P}R + Q\overline{P}$$



The circuit acts as a

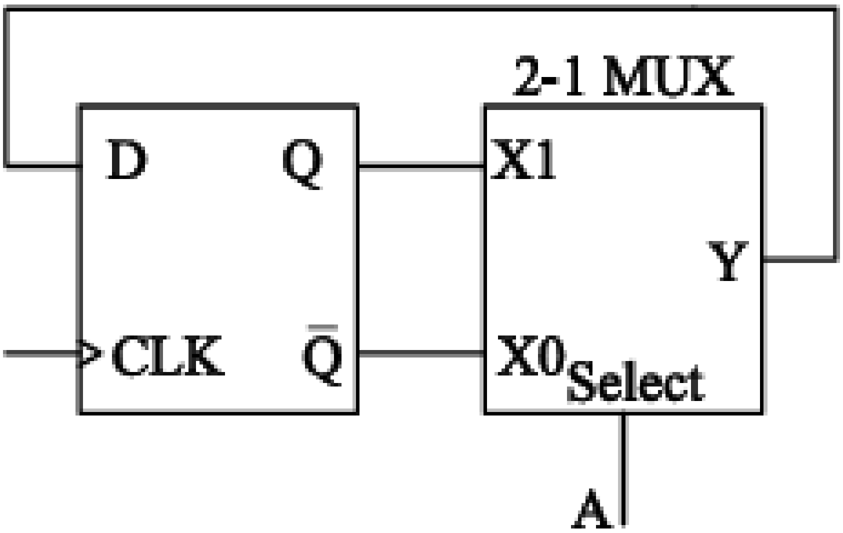
(A) 4 bit adder giving P + Q

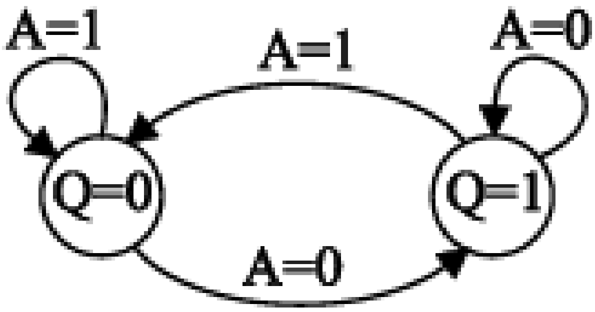
(B) 4 bit subtractor giving P – Q

(C) 4 bit subtractor giving Q – P

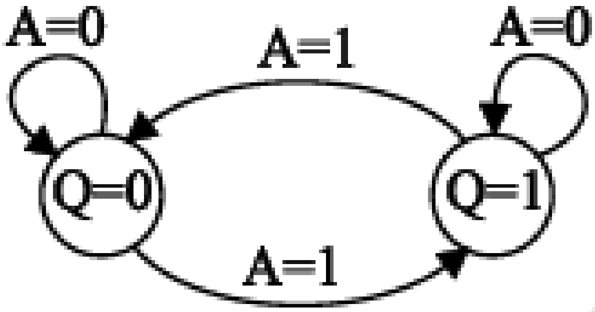
(D) 4 bit adder giving P + Q + R

(1) The state transition diagram for the circuit shown is :

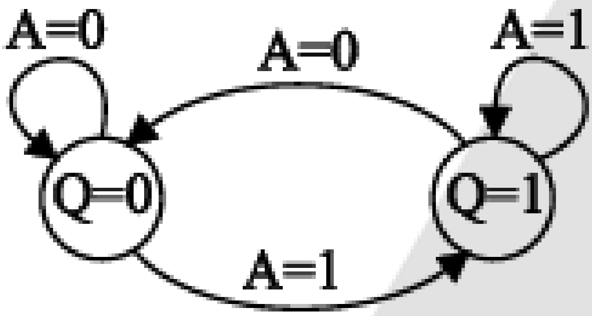




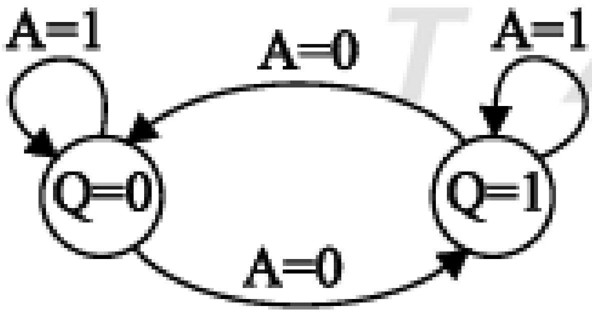
(A)



(B)

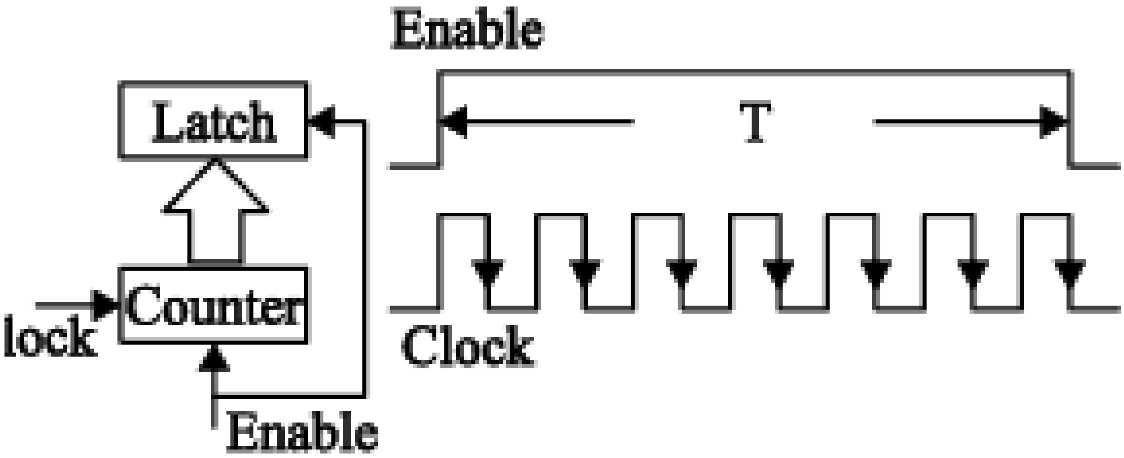


(C)



(D)

) The pulse width T of an asynchronous pulse is measured by a counter with an edge-triggered clock of known frequency f_c as shown in the figure below :



The pulse width to be measured is applied to the Enable pin of the counter. The counter counts while the Enable is high and is held reset to zero otherwise. The counter output is latched by the negative edge of the Enable signal. The measured pulse width is taken to be N times the clock period, where N is the count reached at the end of a count cycle. Assuming no overflow, the measurement error will be limited to x% of T if

(A) $T > \frac{100}{xf_c}$

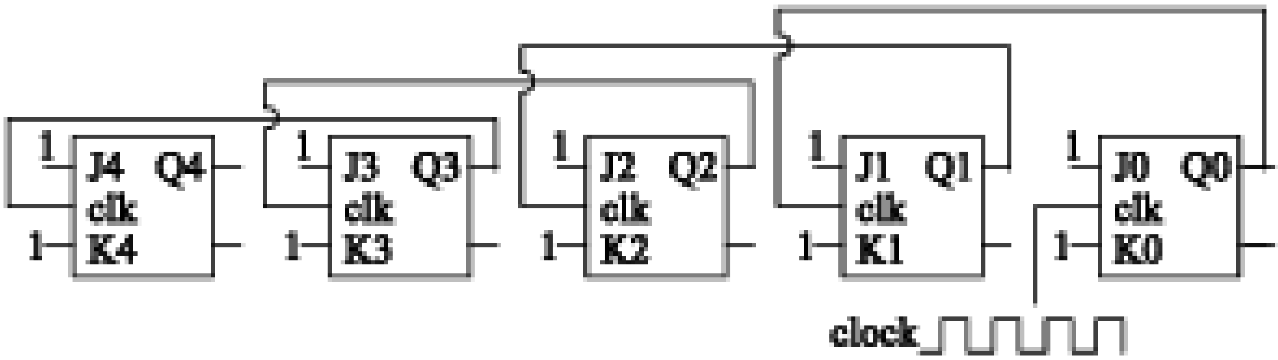
(B) $T < \frac{100}{xf_c}$

(C) $T > \frac{200}{xf_c}$

(D) $T < \frac{200}{xf_c}$

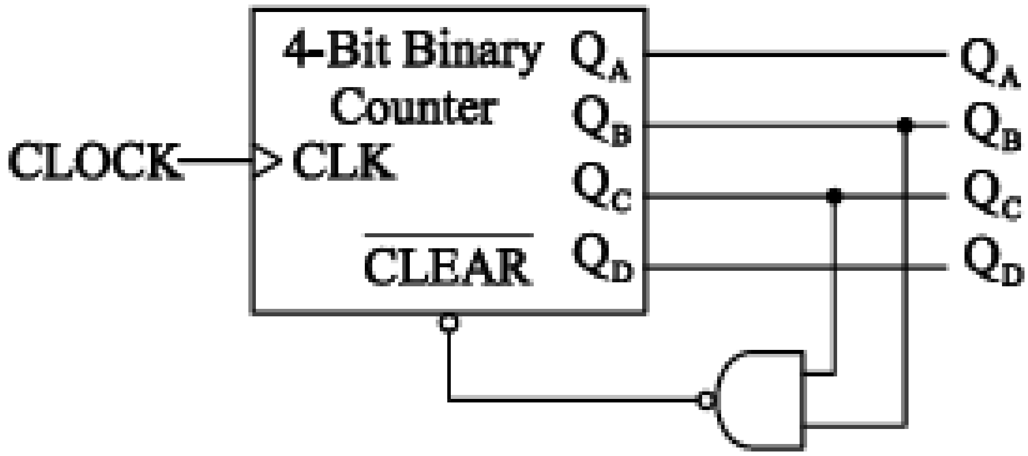
[GATE – EC – 2014]

(2) Five JK flip flops are cascaded to form the circuit shown in figure clock pulses at a frequency of 1 MHz are applied as shown the frequency (in KHz) of the waveform at Q_3 is ----------



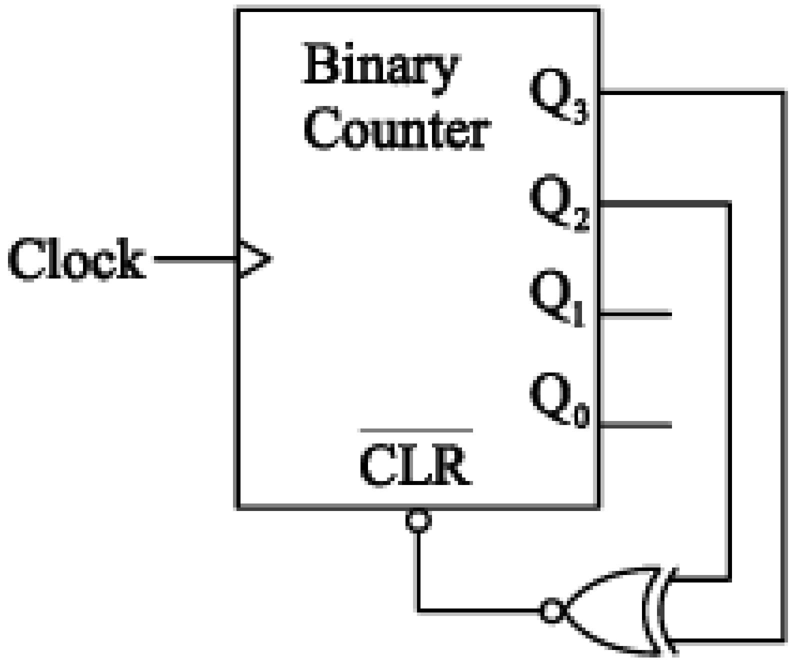
[GATE – EC – 2015]

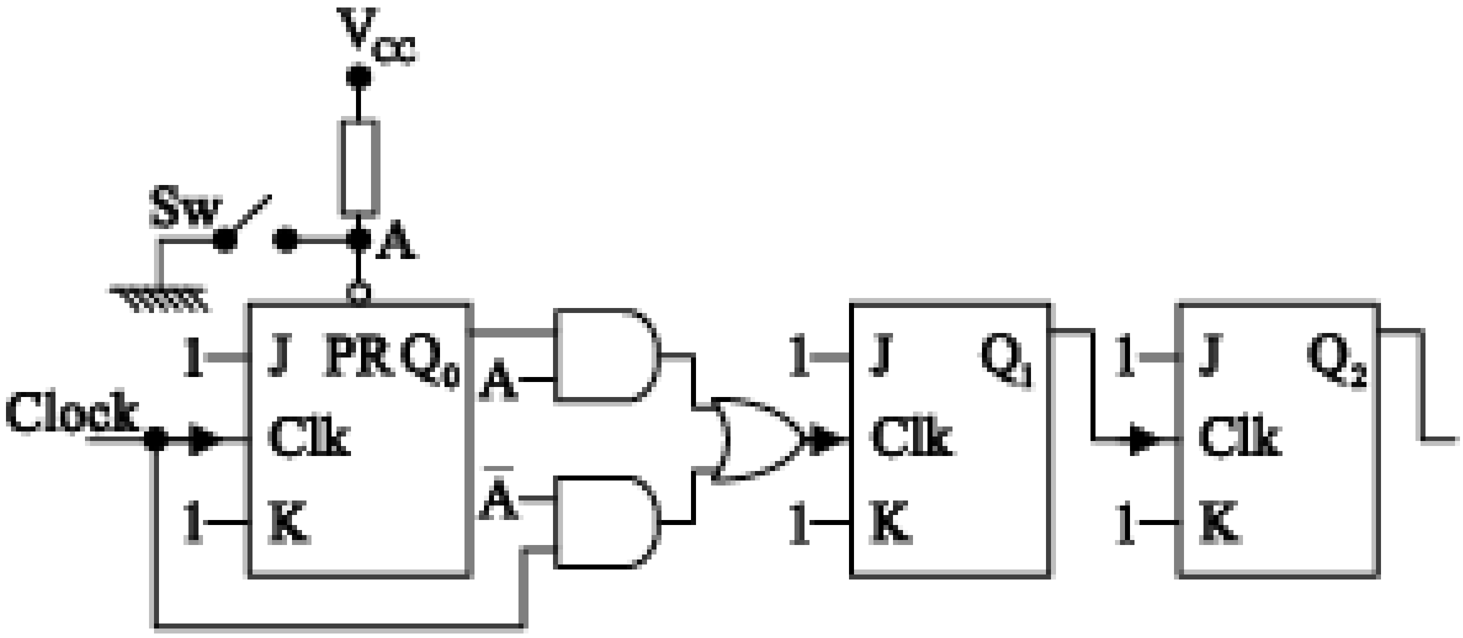
(8) A mod-n counter using a synchronous binary up-counter with synchronous clear input is shown in the figure. The value of n is ____ .



[GATE – EC2 – 2015]

(9) The figure shows a binary counter with synchronous clear input. With the decoding logic show, the counter works as a





(A) Counter outputs are both even and odd numbers

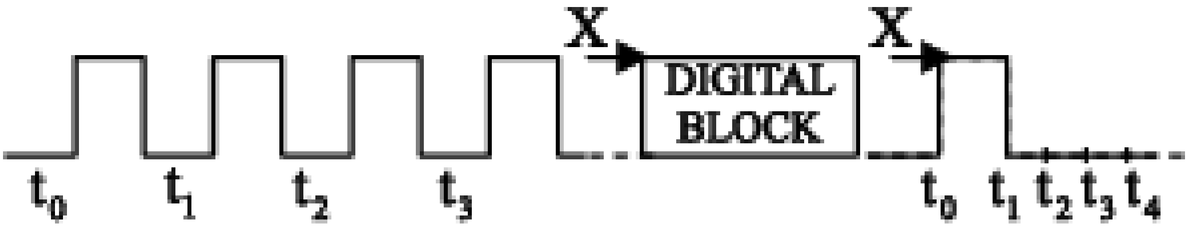
(B) Counter outputs are only odd numbers

(C) Counter outputs are only even numbers

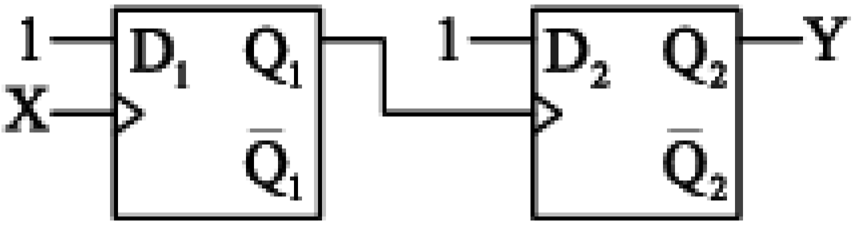
(D) Counter stops counting

[GATE -EC - 2001]

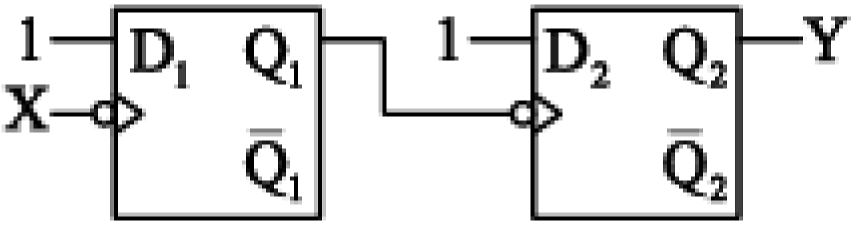
(49) The digital block in the figure is realized using two positive edge triggered D flip – flops. Assume that for $t < t_0$, $Q_1 = Q_2 = 0$. The circuit in the digital block is given by



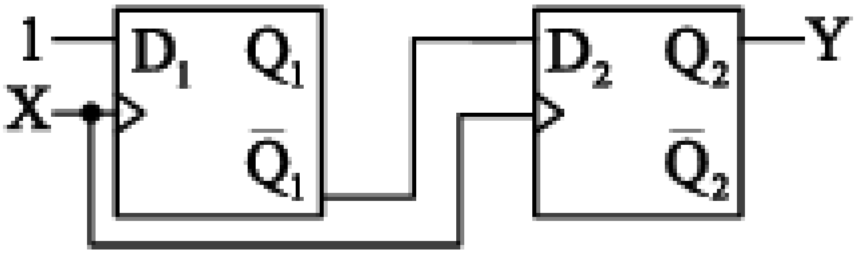
(A)



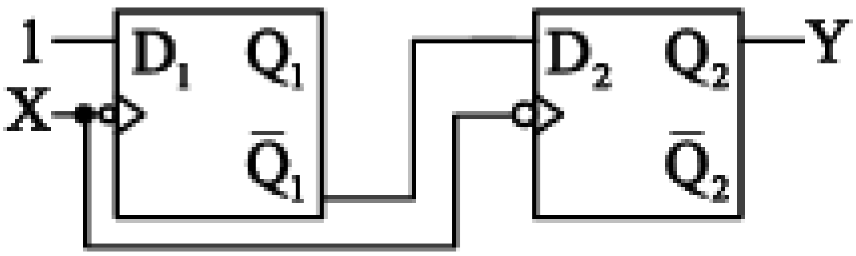
(B)



(C)



(D)



[GATE -EC - 2003]

[GATE -EC - 2011]

The output of a 3-stage Johnson (twisted – ring) counter is fed to a digital – to – analog (D/A) converter as shown in the figure below. Assume all states of the counter to be unset initially. The waveform which represents the D/A converter output V_0 is :